# Curriculum Vitae

NICOLAS LEENAERTS, M.D., Ph.D.

Psychiatrist / Postdoctoral Researcher
UPC KU Leuven/ Harvard University
e-mail:

- nicolas.leenaerts@upckuleuven.be
- nicolasleenaerts@fas.harvard.edu

## EDUCATION

| | |
|---|---|
| 2018-2025 | **Master after Master in Specialized Medicine**<br>Faculty of Medicine, KU Leuven, Belgium |
| 2022-ongoing | **Postgraduate Degree in Cognitive Behavioral Therapy**<br>Faculty of Psychology, KU Leuven, Belgium |
| 2019-2023 | **Doctor of Philosophy in Biomedical Sciences**<br>Cognitive and Molecular Neurosciences, Faculty of Medicine, KU Leuven, Belgium |
| 2014-2018 | **Master of Science in Medicine (Magna Cum Laude)**<br>Faculty of Medicine, KU Leuven, Belgium |
| 2011-2014 | **Bachelor of Science in Medicine (Summa Cum Laude)**<br>Faculty of Medicine, KU Leuven, Belgium |

## ADDITIONAL TRAINING

| | |
|---|---|
| 2026 | **Methods for Adapting and Personalizing Cancer Control Interventions**<br>University of Michigan |
| 2025 | **Generative AI for graduate students and postdocs**<br>Harvard University |
| 2025 | **Building a Transformer from Scratch**<br>Harvard University |
| 2024 | **Reinforcement Learning Specialization**<br>University of Alberta (Online) |
| 2022 | **Multivariate Statistics**<br>KU Leuven |
| 2022 | **Writing Course**<br>Nature Publishing (Online) |
| 2021 | **Eastern European Machine Learning Summer School**<br>EEML (Online) |

| | |
|---|---|
| 2021 | **4th Modelling Symposium: Introducing Deep Neural Networks**<br>NeuroWissenschaftiche Gesellschaft (Online) |
| 2021 | **Longitudinal Data Analysis**<br>KU Leuven |
| 2021 | **Introduction to Clustering Analysis**<br>KU Leuven |
| 2021 | **Introduction to Social Network Analysis**<br>FLAMES |
| 2020 | **Summer School on Affective Neuroscience**<br>University of Maastricht |
| 2020 | **Data Transformations**<br>FLAMES |
| 2020 | **Missing Data Methodology**<br>FLAMES |
| 2020 | **Concepts of Multilevel, Longitudinal and Mixed Models**<br>KU Leuven |
| 2019 | **Statistical Parametric Mapping for fMRI and PET**<br>University College London |
| 2019 | **Python for Everybody**<br>University of Michigan (Online) |

# ACADEMIC APPOINTMENTS

| | |
|---|---|
| 2025-Ongoing | **Postdoctoral researcher (USA)**<br>Statistical reinforcement Learning Lab, Harvard University<br>Responsible Faculty: Prof. Dr. Susan Murphy |
| 2023-2025 | **Voluntary researcher (Belgium)**<br>Faculty of Medicine, KU Leuven<br>Responsible Faculty: Prof. Dr. Elske Vrieze |
| 2019-2023 | **PhD Candidate (Belgium)**<br>Faculty of Medicine, KU Leuven<br>Responsible Faculty: Prof. Dr. Elske Vrieze, Dr. Ir. Jenny Ceccarini, Prof. Dr. Stefan Sunaert |
| 2022 | **Visiting Student Researcher (USA)**<br>Idiographic Dynamics Lab, Department of Psychology, University of California, Berkeley<br>Responsible Faculty: Prof. Dr. Aaron Fisher |
| 2015-2016 | **Student Researcher (Belgium)**<br>Faculty of Medicine, KU Leuven<br>Responsible Faculty: Pascal Delheye |

# CLINICAL APPOINTMENTS

| | |
|---|---|
| 2024-2025 | **Psychiatry Resident (Belgium)**<br>UPC KU Leuven: Franciscus |
| 2023-2024 | **Psychiatry Resident (Belgium)**<br>Zorggroep Sint-Kamillus: iFOR, FRB, FRC<br>Strafinrichting Merksplas |
| 2023-2023 | **Psychiatry Resident (Belgium)**<br>OPZ Geel: Opname-eenheid 1<br>Netwerk GGZ Kempen: CKB-team |
| 2018-2023 | **Psychiatry Resident (Belgium)**<br>UPC KU Leuven: Angst & Depressie, ADHD/Atmosfeer, Perinatale psychiatrie, Eetstoornissen |

# OTHER APPOINTMENTS

| | | |
|---|---|---|
| Ad Hoc Reviewer | Appetite<br>Clinical Psychological Science<br>European Eating Disorders Review<br>eLife<br>Frontiers in Psychology<br>International Journal of Eating Disorders<br>Journal of Eating Disorders<br>Journal of Nuclear Medicine<br>Molecular psychiatry | Psychology of Addictive Behaviors<br>Psychological Medicine<br>Translational Psychiatry<br>Transactions on Computer-Human Interaction |

# HONORS, AWARDS, AND SUPPORT

| | |
|---|---|
| 2026 | **MAPCI Fellow**<br>University of Michigan |
| 2025 | **UZ Leuven Future Fund**<br>UPC KU Leuven |
| 2025 | **Postdoctoral Fellowship**<br>Belgian American Educational Foundation |
| 2025 | **Prijs Prof. Dr. Bernard Sabbe**<br>Koninklijke Academie voor Geneeskunde |
| 2025 | **EFPT Porto Research Award**<br>European Federation of Psychiatric Trainees |
| 2023 | **Grant for participating in a conference abroad**<br>Fonds Wetenschappelijk Onderzoek |
| 2022 | **Grant for participating in a conference abroad**<br>Fonds Wetenschappelijk Onderzoek |

| | |
|---|---|
| 2022 | **Grant for a long stay abroad**<br>Fonds Wetenschappelijk Onderzoek |
| 2022 | **Omkadering Jonge Onderzoekers**<br>The Flemish Government |
| 2021 | **Omkadering Jonge Onderzoekers**<br>The Flemish Government |
| 2021 | **Poster Prize**<br>European College Of Neuropsychopharmacology |
| 2016 | **Erasmus+ Grant**<br>European commission |

# PUBLICATIONS

1. Crotti, M., Ben Itzhak, N., Mailleux, L., Kleeren, L., Decraene, L., **Leenaerts, N.,** Lubián-Gutiérrez, M., Feys, H., & Ortibus, E. (2026). Seeing the Unseen: The Neurodevelopmental Factors Related to Visual Impairments in Children With Unilateral Cerebral Palsy. Pediatric neurology, 178, 15–25. Advance online publication. https://doi.org/10.1016/j.pediatrneurol.2026.01.023

2. **Leenaerts N**, Ceccarini J, Weygandt M, Sunaert S, Vrieze E. The effect of stress on delay discounting in female patients with early-onset bulimia nervosa and alcohol use disorder: functional magnetic resonance imaging study. BJPsych Open. 2025 Sep 12;11(5):e207. doi: 10.1192/bjo.2025.10821. PMID: 40937509; PMCID: PMC12451539**.**

3. **Leenaerts N**, Broeders TAA, Ceccarini J, Sunaert S, Schoonheim MM, Vinkers CH, Vrieze E. Dynamic reconfigurations of brain subnetworks in female patients with alcohol use disorder or bulimia nervosa: a resting-state functional magnetic resonance imaging study. Eur Arch Psychiatry Clin Neurosci. 2025 Jul 3. doi: 10.1007/s00406-025-02055-3. Epub ahead of print. PMID: 40608112.

4. Hagan K, **Leenaerts N**, Walsh BT, Ranzenhofer L. Applying machine learning to ecological momentary assessment data to identify predictors of loss-of-control eating and overeating severity in adolescents: A preliminary investigation. Appetite. 2025 Feb 6;207:107900. doi: 10.1016/j.appet.2025.107900. Epub ahead of print. PMID: 39922228.

5. **Leenaerts, N.,** Vaessen, T., Sunaert, S., Ceccarini, J., & Vrieze, E. (2024). Affective dynamics surrounding craving, non-heavy alcohol use and binge drinking in female patients with alcohol use disorder and controls: An experience sampling method study. Addiction (Abingdon, England), 10.1111/add.16682. Advance online publication. https://doi.org/10.1111/add.16682

6. Crotti, M., Ortibus, E., Ben Itzhak, N., Kleeren, L., Decraene, L., **Leenaerts, N.**, Feys, H., & Mailleux, L. (2024). The relation between visual functions, functional vision, and bimanual function in children with unilateral cerebral palsy. Research in developmental disabilities, 152, 104792. Advance online publication. https://doi.org/10.1016/j.ridd.2024.104792

7. **Leenaerts, N.**, Ceccarini, J., Sunaert, S., & Vrieze, E. (2024). The relation between stress-induced dopamine release in the ventromedial prefrontal cortex, fronto-striatal functional connectivity, and negative urgency: A multimodal investigation using [18F]Fallypride PET, MRI and experience sampling. Behavioural brain research, 115138. Advance online publication. https://doi.org/10.1016/j.bbr.2024.115138

8. **Leenaerts, N.**, Soyster, P., Ceccarini, J., Sunaert, S., Fisher, A., & Vrieze, E. (2024). Person-specific and pooled prediction models for binge eating, alcohol use and binge drinking in bulimia nervosa and alcohol use disorder. Psychological medicine, 1–16. Advance online publication. https://doi.org/10.1017/S0033291724000862

9. Vrieze, E., & **Leenaerts, N.** (2023). Neuronal activity and reward processing in relation to binge eating. Current opinion in psychiatry, 36(6), 443–448. https://doi.org/10.1097/YCO.0000000000000895

10. Schroyen, G., Sleurs, C., Ottenbourghs, T., **Leenaerts, N.**, Nevelsteen, I., Melis, M., Smeets, A., Deprez, S., & Sunaert, S. (2023). Changes in leukoencephalopathy and serum neurofilament after (neo)adjuvant chemotherapy for breast cancer. Translational Oncology, 37, 101769. https://doi.org/10.1016/J.TRANON.2023.101769

11. Melis, M., Schroyen, G., Blommaert, J., **Leenaerts, N.**, Smeets, A., Van Der Gucht, K., Sunaert, S., & Deprez, S. (2023). The Impact of Mindfulness on Functional Brain Connectivity and Peripheral Inflammation in Breast Cancer Survivors with Cognitive Complaints. Cancers, 15(14), 3632. https://doi.org/10.3390/cancers15143632

12. Melis, M., Schroyen, G., **Leenaerts, N.**, Smeets, A., Sunaert, S., Van der Gucht, K., Deprez, S. with Melis, M. (2023). The impact of mindfulness on cancer-related cognitive impairment in breast cancer survivors with cognitive complaints. Cancer, 129 (7), 1105-1116. doi: 10.1002/cncr.34640

13. **Leenaerts, N.**, Vaessen, T., Sunaert, S., Ceccarini, J., Vrieze, E. with Leenaerts, N. (2023). How Negative Affect Does and Does Not Lead to Binge Eating-The Importance of Craving and Negative Urgency in Bulimia Nervosa. Journal Of Psychopathology And Clinical Science. doi: 10.1037/abn0000830

14. Schroyen G, Schramm G, Van Weehaeghe D, **Leenaerts N**, Vande Casteele T, Blommaert J, Koole M, Smeets A, Van Laere K, Sunaert S and Deprez S (2022) Cerebral glucose changes after chemotherapy and their relation to long-term cognitive complaints and fatigue. Front. Oncol. 12:1021615. doi:10.3389/fonc.2022.1021615

15. Radwan, A., Decraene, L., Dupont, P., **Leenaerts, N.**, Simon-Martinez, C., Klingels, K., Ortibus, E., Feys, H., Sunaert, S., Blommaert, J., Mailleux, L. with Blommaert, J. (joint last author), Mailleux, L. (joint last author), Blommaert, J. (2023). Exploring structural connectomes in children with unilateral cerebral palsy using graph theory. Human Brain Mapping. doi: 10.1002/hbm.26241

16. **Leenaerts, N.**, Jongen, D., Ceccarini, J., Van Oudenhove, L., & Vrieze, E. (2022). The neurobiological reward system and binge eating: A critical systematic review of neuroimaging studies. International Journal Of Eating Disorders, 38 pages. doi:10.1002/eat.23776

17. **Leenaerts, N.**, Ceccarini, J., Sunaert, S., & Vrieze, E. (2022). The impact of COVID-19-related smell and taste disorders on a patient with bulimia nervosa: a case report. Neurocase, 28(1), 72-76. doi:10.1080/13554794.2021.2024859

18. Demyttenaere, K., **Leenaerts, N.**, Acsai, K., Sebe, B., Laszlovszky, I., Barabassy, A., . . . Correll, C. U. (2021). Disentangling the symptoms of schizophrenia: Network analysis in acute phase patients and in patients with predominant negative symptoms. European Psychiatry, 65(1), 10 pages. doi:10.1192/j.eurpsy.2021.2241

19. **Leenaerts, N.**, Vaessen, T., Ceccarini, J., & Vrieze, E. (2021). How COVID-19 lockdown measures could impact patients with bulimia nervosa: Exploratory results from an ongoing experience sampling method study. Eating Behaviors, 41, 1-6. doi:10.1016/j.eatbeh.2021.101505

# PREPRINTS

1. Fisher, A. J., **Leenaerts, N.,** & Victor, S. E. (2023, September 16). A Logical Approach to Necessity and Sufficiency. https://doi.org/10.31234/osf.io/jt7m4

# ABSTRACTS, PRESENTATIONS, AND POSTERS

1. **Leenaerts N.**, Goldstein S., Coughlin L., Thomas J. G., Wing R.R., Zhang F., Klasnja P., Hoover A., Murphy S. (2025). Improving Just-In-Time Adaptive Interventions for Overweight and Obesity. Technology In Psychiatry Summit. Online.

2. De Schuyteneer E., **Leenaerts N.**, Vrieze E., Meule A., Kolar D., Wuttke E., Voderholzer U. (2025). Developing a clinical prediction model for persistent depressive symptoms after treatment of eating disorders. Eating Disorders Research Society. Newport Beach, USA.

3. **Leenaerts, N.**, Broeders, T., ceccarini, j., Sunaert, S., Schoonheim, M., Vinkers, C., & Vrieze, E. (2024). Dynamic reconfigurations of brain subnetworks in alcohol use disorder or bulimia nervosa: A resting-state functional magnetic resonance imaging study. Eating Disorders Research Society. Sitges, Spain.

4. **Leenaerts, N.**, Vrieze, E., Ceccarini, J., & Sunaert, S. (2024). Spinning Out Of Control: How Stress and Negative Affect Lead to Binge Behavior in Bulimia Nervosa and Alcohol Use Disorder. Psychiatry Research Day. Leuven, Belgium.

5. **Leenaerts, N.**, Vaessen, T., Sunaert, S., Ceccarini, J., & Vrieze, E. (2023). Person-specific and pooled prediction models for binge eating, alcohol use and binge drinking in bulimia nervosa and alcohol use disorder. In Eating Disorders Research Society. Boston, United States of America.

6. **Leenaerts, N.**, Vaessen, T., Sunaert, S., Ceccarini, J., & Vrieze, E. (2022). HOW NEGATIVE AFFECT LEADS TO BINGE EATING: THE IMPORTANCE OF IMPULSIVITY AND CRAVING IN BULIMIA NERVOSA. In Eating Disorders Research Society. Philadelphia, United States of America.

7. **Leenaerts, N.**, Ceccarini, J., Sunaert, S., & Vrieze, E. (2021). Temporal dynamics of impulsivity and craving before and after a binge drinking episode: results from an ongoing experience sampling study. In ESBRA 2021. Timisoara, Romania.

8. **Leenaerts, N.**, Ceccarini, J., Sunaert, S., & Vrieze, E. (2021). Striatal cerebral blood flow changes in patients with recent-onset bulimia nervosa and alcohol use disorder. In ECNP 2021. Lisbon, Portugal.

9. **Leenaerts, N.**, Ceccarini, J., Sunaert, S., & Vrieze, E. (2021). Temporal dynamics of impulsivity and craving before and after a binge eating episode: results from an ongoing experience sampling study. In EDRS 2021. Online.

10. **Leenaerts, N.**, Vaessen, T., Ceccarini, J., & Vrieze, E. (2021). How COVID-19 lockdown measures could impact patients with bulimia nervosa: Exploratory results from an ongoing experience sampling method study. In European Psychiatric Association. Online.

11. **Leenaerts, N.**, Vaessen, T., Ceccarini, J., & Vrieze, E. (2020). Beloningsgevoeligheid bij patiënten met bulimia nervosa: voorlopige resultaten van een experience sampling method studie. In Nationale Academie Eetstoornissen. Beesd.

12. **Leenaerts, N.**, Vaessen, T., Ceccarini, J., & Vrieze, E. (2020). How COVID-19 lockdown measures could impact patients with bulimia nervosa: exploratory results from an ongoing experience sampling method study.. In VAE Congres 2020 'Andere Perspectieven'. Online.

13. **Leenaerts, N.**, Vaessen, T., Ceccarini, J., & Vrieze, E. (2020). Linking stress to impulsivity in recent-onset bulimia nervosa and alcohol use disorder: preliminary results from an ecological momentary assessments study. In European College of Neuropsychopharmacology. Vienna.

14. **Leenaerts, N.**, Vrieze, E., Sunaert, S., Van Laere, K., & Ceccarini, J. (2020). Effects of lifetime alcohol consumption on surface morphometry in alcohol-dependent patients. In Organization for Human Brain Mapping. Montreal.

# SOFTWARE

1. https://github.com/nicolasleenaerts/NLML: A GitHub repository with wrappers that allow researchers to build methodologically valid pooled and person-specific elastic net regression models to predict daily life behavior.

2. https://github.com/nicolasleenaerts/NLMLapp: An easy to use R Shiny app that allows researchers to analyze their data with the wrappers from the NLML GitHub repository.

3. https://github.com/nicolasleenaerts/setweaver: R package designed to help users create sets of variables based on a mutual information approach

# PRESS

1. https://nicolasleenaerts.github.io/press/: An overview of all instances my work appeared in the press.

# MENTORING

| | |
|---|---|
| 2025-Ongoing | **Edward Peuskens** (Role: Co-promotor)<br>Master of Science in Medicine, KU Leuven |
| 2025-Ongoing | **Hannah Bogaerts** (Role: Co-promotor)<br>Master of Science in Medicine, KU Leuven |
| 2022-2023 | **Ellen Boon** (Role: Daily Supervisor)<br>Master of Science in Biomedical Sciences, KU Leuven<br>Thesis: The effect of stress on interoception in patients with Anorexia Nervosa. |
| 2021-2022 | **Lotte Buyle** (Role: Daily Supervisor)<br>Master of Science in Biomedical Sciences, KU Leuven<br>Thesis: Reward functioning in patients with Bulimia Nervosa or an Alcohol Use Disorder: Looking at daily life. |
| 2020-2021 | **Maxime Voet** (Role: Daily Supervisor)<br>Master of Science in Biomedical Sciences, KU Leuven<br>Thesis: Stress reactivity is related to higher dopamine release in the prefrontal and cingulate cortex. |
| 2020-2021 | **Danny Majid** (Role: Daily Supervisor)<br>Master of Science in Medicine, Linköping university, Sweden<br>Thesis: Stress and delay discounting in patients with bulimia nervosa: a functional magnetic resonance study |

# OTHER ACTIVITIES

| | |
|---|---|
| 2024-2025 | **Vlaamse Vereniging Assistenten Psychiatrie**<br>Organization representing psychiatry residents in Flanders (Role: Member) |
| 2024-2025 | **Student Representative in UPC KU Leuven campus Kortenberg** |
| 2022 | **Psychiatry 2.0: Towards a new way of thinking**<br>Inter-universitary seminar (Role: Organizer) |
| 2021 | **Machine Learning in Psychiatry for Dummies**<br>Inter-universitary seminar (Role: Organizer) |
| 2021-Ongoing | **Current Topics in Psychiatry**<br>Journal Club (Role: Co-founder) |
| 2015 | **Summer School of Psychiatry**<br>Summer school (Role: Organizer) |

# COMMUNITY SERVICE

| | |
|---|---|
| 2023-ongoing | **Te Gek**<br>Charity aiming to break the stigma around mental health (Role: Volunteer) |
| 2021-2023 | **Bijleshuis**<br>Company offering tutoring services (Role: Tutor of statistics and medical writing) |

| | |
|---|---|
| 2014 - 2016 | **Medica**<br>Student organization (Role: Member, Chief Editor of the ‘Akuut’ journal, volunteer for the ‘Warme Landen’ charity) |
| 2003-Ongoing | **Koninklijke Harmonie “De Heren van Zichem”**<br>Harmonic Orchestra (Role: Saxophone player, co-organizer of local events) |